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Prioritization of Risks in Agile Software Projects Through an Analytic Hierarchy Process Approach

Ayesha Ziana M[a], Charles J[b,*]

[a]*Department of Computer Science, Noorul Islam Centre for Higher Education, Kumaracoil-629180, Tamil Nadu, India*
[b]*Department of Software Engineering, Noorul Islam Centre for Higher Education, Kumaracoil-629180, Tamil Nadu, India*
jebapillaicharles@gmail.com

Abstract

Risk management is often overlooked in the software industry, especially in agile methodology-based projects, which can have negative impacts on the schedule, budget, and quality of a project, eventually leading to customer dissatisfaction. Hence, it is essential to identify the risks and prioritize them based on their level of importance. This study has utilized the Analytic Hierarchy Process (AHP) methodology for prioritizing the software risk factors identified during our previously executed risk identification phase experimental study in small and medium software enterprises that follow agile and traditional methodologies for software development. Among the 16 risk factors categorized under four different risk natures named 'difficulties', 'changes', 'mistakes', and 'challenges', the 'challenges' risk nature category is found to be the most significant one by the research findings, followed by 'difficulties', 'mistakes', and 'changes' risk natures. The results also highlight that 'proper utilization of funds', 'complicated tasks', and 'convincing the customer with alternative technologies' are the most significant risk factors. The study gives an insight into the possible risk factors in a software environment, along with their significance.

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Keywords: Agile Software Development (ASD); Multi-criteria decision-making; Risk prioritization; Analytic Hierarchy Process (AHP); Software risk assessment; Risk nature

1. Introduction

Recently, Agile Software Development (ASD) has become more popular in the software industry since it improves the speed of the development process and has greater acceptance of changing requirements in dynamic environments [1]. Agile projects require a meticulous approach for characterizing, evaluating, and managing risks due to the fact that there are chances for risks with high negative effects to be ignored, thus leading to problems in decision-making. This drawback of Agile derives from the conclusion that most of the failures faced by Agile projects are because

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of ineffectiveness in assessing and managing risks, and hence it is important to develop structured risk models and techniques [2].

According to the famous Project Management Body of Knowledge (PMBOK) standard, risk management is one of the most vital areas of assistance in project management. Structured risk management is concerned with the identification, classification, analysis, and mitigation of risks. Risk assessment is one of the two-stage processes involved in risk management and includes the identification, analysis, and prioritization of risks [3]. It is essential to group and prioritize the extensive list of risks obtained to provide a clear-cut representation of which of the risks need to be handled and allocate the resources accordingly. As per the ISO 12207 standard, which describes the processes carried out in the Software Development Life Cycle (SDLC), an objective can be related from several points of view, some of which include budget-related, health-related, safety-related, and environmental. There are chances for risk to occur at various levels of the enterprise, resulting from a wide range of both internal and external factors, which can have great control over the likelihood of a risk and its effects on the business objectives [1].

Usually, the practitioners of a project identify the risks, which are then evaluated and assessed to implement the appropriate risk mitigation strategies. To improve the evaluation process, a prioritization technique is essential. Thus, the main contribution of the study is focused on prioritizing the previously identified risks through an Analytic Hierarchy Process (AHP) methodology.

2. Related work

In [4], a metaheuristic approach based on the Particle Swarm Optimization algorithm and an Agile-based Risk Rank methodology were proposed for the prioritization and optimization of the 45 identified risks from the literature. The experiments showed significantly better results than the previous approaches.

The AHP process acted as an entrance to finding the risky requirements, according to [5]. The experimental case study provided a positive outcome with a Consistency Ratio (CR) value less than 0.1, thus indicating the consistency of the approach.

The authors of [6] have made a significant contribution by proposing a few measures of importance for risk prioritization, namely ‘Risk Achievement Importance’, ‘Risk Reduction Importance’, ‘Marginal Risk Importance’, and ‘Marginal Risk Interaction Importance’, which were then implemented using a simulation-based risk network model.

A novel FMEA-based model for prioritizing risks is illustrated in [7], which uses two functions named the Risk Isosurface (RI) function and the Risk Priority Index (RPI) function to prioritize the failure modes.

Another hybrid approach using the Intuitionistic Fuzzy-based TOPSIS (IF-TOPSIS), adaptive neuro-fuzzy inference system-based Multi Criteria Decision Making (ANFIS MCDM), and Fuzzy Decision-MAking Trial and Evaluation Laboratory (Fuzzy DEMATEL) methodologies was proposed in [8] and provided a clear significance and ranking of the software risk factors. The Enhanced Crow Search Algorithm (ECSA) was used to adjust the ANFIS parameters for better accuracy. The framework aimed to assist the software experts in handling the risks earlier.

An Analytic Hierarchy Process (AHP)-based software risk prioritization model was proposed in [9], which was further evaluated by the Partial Least Square (PLS)-Structural Equation Modeling (SEM) in terms of usability attributes such as efficiency, effectiveness, user satisfaction, learnability, accessibility, navigation, content availability, structure and design, and model acceptance.

A risk matrix-based Monte Carlo Simulation was introduced in [10], where a few significant risk metrics, including ‘Risk signature’, were proposed in building the model for prioritizing risks.

The MOORA (Multi-Objective Optimization on the basis of Ratio Analysis) played a major role in assessing and prioritizing Occupational Health and Safety (OHS) risks in the hybrid model proposed in [11]. The method made use of the outputs from the Failure Mode and Effect Analysis (FMEA) decision matrix and Fuzzy Cognitive Map (FCM) as its input.

Another hybrid model with the FMEA and Dual Hesitant Fuzzy Sets (DHFS) was illustrated in [12], which is useful in situations where judgements are made out of hesitations and hence applicable in different contexts of decision-making problems.

In [13], an AHP-based approach was also used in prioritizing the challenges faced in distributed agile projects, in which 22 challenges from the literature were used for the study. The AHP method prioritized the challenges along

with the grouped 4 categories, namely 'management', 'team', 'technology', and 'process', depending on their relative importance.

The novel complementary matrix diagram proposed in [14] executes a cumulative manner of risk prioritization in which a critical risk is finalized based on the individual and cumulative grades of criticality. This is in contrast to the traditional Farmer diagram, where only individual criticality is taken into account.

The AHP methodology is applicable in different domains, including risk management [5]. This motivated us to apply the methodology to prioritize the risks identified in software projects in small and medium software enterprises.

3. Methodology

The Analytic Hierarchy Process (AHP) is a multi-criteria decision-making methodology applied in several fields in order to find solutions to complex problems [13]. It consists of the following processing stages:

Step 1: The first and foremost step in this process is to dissolve the complex problem into a hierarchical structure of elements that are interrelated to each other. Table 1 lists the criteria (risk nature) and the sub-criteria (risk factors) used in the study. Fig. 1 illustrates the same, along with the goal of the study, in a hierarchical manner.

Table 1. Risk Factors identified in our previous study

Risk Factor No.	Risk Factor	Risk Nature
R1	Complicated tasks	Difficulties
R2	Complications in finding the right person for task allocation	Difficulties
R3	Difficulty in developing will be faced by the developers, leading to a time delay	Difficulties
R4	Difficulty in adjusting our schedule to that of the customer	Difficulties
R5	Unexpected environmental changes or pandemic situations leading to delays	Changes
R6	Handovers to a new team member due to personnel turnover can be problematic	Changes
R7	Change requests during the sprint execution	Changes
R8	Adaptation of agile methodology in the project if using it for the first time	Changes
R9	Coding is not done according to the design	Mistakes
R10	Improper communication among the clients and the team	Mistakes
R11	Imperfect sprint planning	Mistakes
R12	Mistakes due to poor standards in testing and documentation	Mistakes
R13	The high-investment project may require high-standard techniques and methodologies	Challenges
R14	Proper utilization of funds	Challenges
R15	Scalability of resources	Challenges
R16	Convincing the customer with alternative technologies	Challenges

Step 2: Find the priority vector (weight) of each criterion and sub-criteria using the pairwise comparison. A 9-point comparison scale (given in Table 2) is used for this purpose, which generates a pairwise comparison matrix for each criterion and its corresponding sub-criteria. The normalized matrix is then generated by dividing each element of the matrix by the sum value of the respective column. The priority vector (weight) is then generated by computing the average of the row elements.

Table 2. The 9-point comparison scale description for AHP

Description	Intensity of Importance
Equally important	1
Moderately important	3
Strongly more important	5
Very strongly more important	7
Extremely more important	9
Intermediate values	2,4,6,8

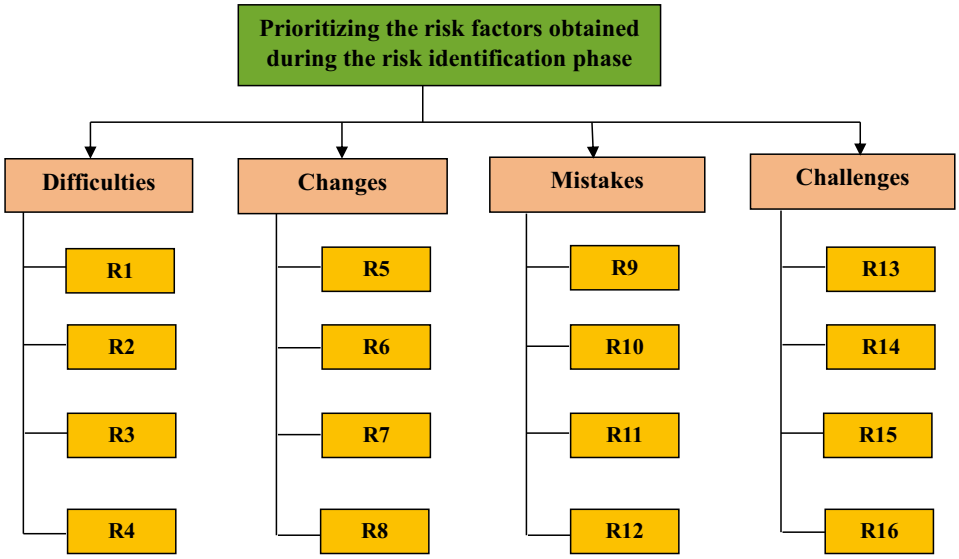


Fig. 1. Hierarchy of the study using Analytic Hierarchy Process methodology

Step 3: The next step is to verify the consistency of the pairwise comparison matrix. To do this, the ratio between the weighted sum values and the priority vector is calculated. The λ_{max} value is then derived by averaging the ratio values. The Consistency Index (CI) and the Consistency Ratio (CR) are found using equations (1) and (2).

$$ConsistencyIndex(CI) = \frac{\lambda_{max} - n}{n - 1} \tag{1}$$

where 'n' is the number of items taken for comparison.

$$ConsistencyRatio(CR) = \frac{CI}{RI} \tag{2}$$

where R.I. refers to the 'Random Index'. 'Random Index' is the consistency index of a randomly generated pairwise matrix. The random consistency index table for up to 10 criteria is shown in Table 3.

Table 3. The Random Consistency Index table

Size of the matrix, n	1	2	3	4	5	6	7	8	9	10
Random Index	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

In our study, we have taken 4 items for comparison every time, and the corresponding random index element is 0.90, as per the above table. With this, we can calculate the Consistency Ratio (CR) by applying it to equation (2). The resultant value must be less than 0.10 for us to conclude that the priority vector of the risk factors can be accepted and that the matrix has enough consistency. Otherwise, the above steps have to be repeated.

Step 4: Based on the priority vector (local weight) calculated in Step 2, the local ranking of the risk factors within each 'risk nature', has to be assigned.

Step 5: The global weight of each risk factor is then computed by multiplying the priority weight of its 'risk nature' with the local weight of the risk factor computed in Step 2. Based on this value, the global ranking for the risk factors needs to be assigned.

4. Proposed Methodology

The main goal of the study is to prioritize the risk factors previously identified by our experimental study in the risk identification phase. Three different groups of software practitioners participated in the experiments, and they identified 435 risks altogether for the given project specifications during the sessions. The risk factors fall under various 'risk nature' categories named 'over-doing', 'under-doing', 'mistakes', 'conceptual problems', 'changes', 'differences', 'difficulties', 'dependency', 'conflicts', 'issues', and 'challenges'. All these risk factors were assessed based on the 'score' and 'criticality level' by experienced experts in the field. Among the 217 unique ideas generated during the experimental sessions, we have selected 16 risk factors randomly, which fall under the 'risk nature' named 'difficulties', 'changes', 'mistakes', and 'challenges'. 'Difficulty' plays a major role in the failure of big projects, according to McKinnie [15]. 'Changes' can occur at any instance throughout the lifecycle of a project and can be the reason for its win or loss [16]. 'Mistakes' arise when the right concepts are wrongly implemented [17]. 'Challenges' pertain to the new circumstances that offer discrepancies or act as a development catalyst [18]. Risks of such nature are expected to occur in software projects, and hence they need to be prioritized for accomplishing the mitigation activities. Three software practitioners with notable experience contributed to generating the pairwise comparisons in this study.

The steps illustrated in the previous section are applied to all the risk factors given in Table 1. The outcomes are further demonstrated in tables 4 to 15.

Table 4. Pairwise Comparison Matrix between risks of 'difficulties' nature

Risk Factor No.	R1	R2	R3	R4
R1	1	7	5	5
R2	1/7(0.143)	1	1/5(0.2)	1/3(0.33)
R3	1/5(0.2)	5	1	3
R4	1/5(0.2)	3	1/3(0.33)	1
Sum of the column	1.543	16	6.53	9.33

Table 5. Normalized matrix for 'difficulties' nature

Risk Factor No.	R1	R2	R3	R4	Priority Vector
R1	0.648	0.4375	0.7657	0.5359	0.5968
R2	0.093	0.0625	0.0306	0.0354	0.0554
R3	0.1296	0.3125	0.1531	0.3215	0.2269
R4	0.1296	0.1875	0.0505	0.1072	0.1187

For 'difficulties' risk nature,

λ_{max} = (4.5452 + 4.065 + 4.3147 + 4.0337)/4 = 4.23965;

Consistency Index (CI) = (4.23965 - 4)/(4 -1) = 0.23965/3 = 0.079883;

Consistency Ratio (CR) = 0.079883/0.90 = 0.08876 < 0.10. So, we can conclude that the priority vector of the risk factors can be accepted.

Similarly, for 'changes' risk nature,

λ_{max} = (4.2179 + 4.1579 + 4.03516 + 4.079)/4 = 4.12249; CI = (4.12249 - 4)/(4 -1) = 0.12249/3 = 0.04083;

CR = 0.04083/0.90 = 0.0454 < 0.10. So, we can conclude that the priority vector of the risk factors can be accepted.

Table 6. Consistency Index and Consistency Ratio Calculation for ‘difficulties’ nature

Risk Factor No.	R1	R2	R3	R4	Weighted sum value	Priority vector	Weighted sum value/Priority vector
R1	0.5968	0.3878	1.1345	0.5935	2.7126	0.5968	4.5452
R2	0.0853	0.0554	0.0454	0.0391	0.2252	0.0554	4.065
R3	0.119	0.277	0.2269	0.3561	0.979	0.2269	4.3147
R4	0.119	0.1662	0.0749	0.1187	0.4788	0.1187	4.0337

Table 7. Pairwise Comparison Matrix between risks of ‘changes’ nature

Risk Factor No.	R5	R6	R7	R8
R5	1	4	4	4
R6	1/4 (0.25)	1	2	2
R7	1/4 (0.25)	1/2 (0.5)	1	2
R8	1/4 (0.25)	1/2 (0.5)	1/2 (0.5)	1
Sum of the column	1.75	6	7.5	9

Table 8. Normalized matrix for ‘changes’ nature

Risk Factor No.	R5	R6	R7	R8	Priority Vector
R5	0.5714	0.667	0.533	0.44	0.55285
R6	0.143	0.167	0.267	0.22	0.19925
R7	0.143	0.083	0.133	0.22	0.14475
R8	0.143	0.083	0.067	0.11	0.10075

Table 9. Consistency Index and Consistency Ratio Calculation for ‘changes’ nature

Risk Factor No.	R5	R6	R7	R8	Weighted sum value	Priority vector	Weighted sum value/Priority vector
R5	0.55285	0.797	0.579	0.403	2.33185	0.55285	4.2179
R6	0.138213	0.19925	0.2895	0.2015	0.828465	0.19925	4.1579
R7	0.138213	0.09963	0.14475	0.2015	0.5840875	0.14475	4.0351
R8	0.138213	0.09963	0.07238	0.10075	0.4109625	0.10075	4.079

Table 10. Pairwise Comparison Matrix between risks of ‘mistakes’ nature

Risk Factor No.	R9	R10	R11	R12
R9	1	1/5 (0.2)	2	2
R10	5	1	5	5
R11	1/2 (0.5)	1/5 (0.2)	1	2
R12	1/2 (0.5)	1/5 (0.2)	1/2 (0.5)	1
Sum of the column	7	1.6	8.5	10

For ‘mistakes’ risk nature,

λ_{max} = (4.1627 + 4.2386 + 4.0095 + 4.0715)/4 = 4.120575;

CI = (4.120575 - 4)/(4 - 1) = 0.120575/3 = 0.04019167;

CR = 0.04019167/0.90 = 0.0447 < 0.10. So, we can conclude that the priority vector of the risk factors can be accepted.

For ‘challenges’ risk nature,

λ_{max} = (4.0095 + 4.2386 + 4.0814 + 4.1627)/4 = 4.12305;

Table 11. Normalized matrix for 'mistakes' nature

Risk Factor No.	R9	R10	R11	R12	Priority Vector
R9	0.143	0.125	0.235	0.2	0.17575
R10	0.714	0.625	0.588	0.5	0.60675
R11	0.071	0.125	0.118	0.2	0.1285
R12	0.071	0.125	0.059	0.1	0.08875

Table 12. Consistency Index and Consistency Ratio Calculation for 'mistakes' nature

Risk Factor No.	R9	R10	R11	R12	Weighted sum value	Priority vector	Weighted sum value/Priority vector
R9	0.17575	0.12135	0.257	0.1775	0.7316	0.17575	4.1627
R10	0.87875	0.60675	0.6425	0.44375	2.57175	0.60675	4.2386
R11	0.087875	0.12135	0.1285	0.1775	0.515225	0.1285	4.0095
R12	0.087875	0.12135	0.06425	0.08875	0.36135	0.08875	4.0715

Table 13. Pairwise Comparison Matrix between risks of 'challenges' nature

Risk Factor No.	R13	R14	R15	R16
R13	1	1/5 (0.2)	2	1/2 (0.5)
R14	5	1	5	5
R15	1/2 (0.5)	1/5 (0.2)	1	1/2 (0.5)
R16	2	1/5 (0.2)	2	1
Sum of the column	8.5	1.6	10	7

Table 14. Normalized matrix for 'challenges' nature

Risk Factor No.	R13	R14	R15	R16	Priority Vector
R13	0.118	0.125	0.2	0.071	0.1285
R14	0.588	0.625	0.5	0.714	0.60675
R15	0.059	0.125	0.1	0.071	0.08875
R16	0.235	0.125	0.2	0.143	0.17575

Table 15. Consistency Index and Consistency Ratio Calculation for 'challenges' nature

Risk Factor No.	R13	R14	R15	R16	Weighted sum value	Priority vector	Weighted sum value/Priority vector
R13	0.1285	0.12135	0.1775	0.087875	0.515225	0.1285	4.0095
R14	0.6425	0.60675	0.44375	0.87875	2.57175	0.60675	4.2386
R15	0.06425	0.12135	0.08875	0.087875	0.362225	0.08875	4.0814
R16	0.257	0.12135	0.1775	0.17575	0.7316	0.17575	4.1627

CI = (4.12305 - 4)/(4-1) = 0.12305/3 = 0.04102;

CR = 0.04102/0.90 = 0.0456 < 0.10. So, we can conclude that the priority vector of the risk factors can be accepted.

The priority vector (weight) of the risk natures is computed in a similar fashion and tabulated from Tables 16 to 18.

For 'risk nature' categories,

λ_{max} = (4.34 + 4.07 + 4.08 + 4.27)/4 = 16.76/4 = 4.19;

CI= (4.19 - 4)/(4 - 1) = 0.19/3 = 0.063; CR = 0.063/0.90 = 0.07 < 0.10. So, we can conclude that the priority vector of the different risk natures can be accepted.

Table 16. Pairwise Comparison Matrix between different 'risk nature' categories

Risk Nature	Difficulties	Changes	Mistakes	Challenges
Difficulties	1	4	3	1/3 (0.33)
Changes	1/4 (0.25)	1	1/4(0.25)	1/7 (0.143)
Mistakes	1/3 (0.33)	4	1	1/5 (0.2)
Challenges	3	7	5	1
Sum of the column	4.58	16	9.25	1.673

Table 17. Normalized matrix for 'risk nature' categories

Risk Nature	Difficulties	Changes	Mistakes	Challenges	Priority Vector
Difficulties	0.218	0.25	0.324	0.197	0.24725
Changes	0.055	0.0625	0.027	0.085	0.057375
Mistakes	0.072	0.25	0.108	0.12	0.1375
Challenges	0.66	0.4375	0.541	0.6	0.56

Table 18. Consistency Index and Consistency Ratio Calculation for 'risk nature' categories

Risk Nature	Dif.	Ch.	Mis.	Chall.	Weighted sum value	Priority vector	Weighted sum value/Priority vector
Difficulties	0.24725	0.2295	0.4125	0.1848	1.07405	0.24725	4.34
Changes	0.0618125	0.057375	0.034375	0.08008	0.233642625	0.057375	4.07
Mistakes	0.0815925	0.2295	0.1375	0.112	0.5605925	0.1375	4.08
Challenges	0.74175	0.401625	0.6875	0.56	2.390875	0.56	4.27

The local and global weights of the risk factors are shown in Table 19. The local rank is computed on the basis of the priority vector (local weight) of each risk factor. The global weight is the product of the local weight and the corresponding risk-nature weight of each risk factor. Finally, the global rank of the risk factor is assigned on the basis of its global weight.

Table 19. Summary of the local and global rankings of the risk factors

Risk Nature	Weights of the Risk Nature	Risk Factor No.	Local weight	Local rank	Global weight	Global rank
Difficulties	0.24725	R1	0.5968	1	0.1476	2
		R2	0.0554	4	0.0137	12
		R3	0.2269	2	0.0561	6
		R4	0.1187	3	0.0293	9
Changes	0.057375	R5	0.55285	1	0.0317	8
		R6	0.19925	2	0.0114	14
		R7	0.14475	3	0.0083	15
		R8	0.10075	4	0.0058	16
Mistakes	0.1375	R9	0.17575	2	0.0242	10
		R10	0.60675	1	0.0834	4
		R11	0.1285	3	0.0177	11
		R12	0.08875	4	0.0122	13
Challenges	0.56	R13	0.1285	3	0.07196	5
		R14	0.60675	1	0.33978	1
		R15	0.08875	4	0.0497	7
		R16	0.17575	2	0.09842	3

Based on the local and global rankings computed from the above table, the risk factors are ranked and summarized in Table 20.

Table 20. Prioritization of the risk factors

Risk Factor No.	Risk Factor	Priority
R14	Proper utilization of funds	1
R1	Complicated tasks	2
R16	Convincing the customer with alternative technologies	3
R10	Improper communication among the clients and the team	4
R13	The high-investment project may require high-standard techniques and methodologies	5
R3	Difficulty in developing will be faced by the developers, leading to a time delay	6
R15	Scalability of resources	7
R5	Unexpected environmental changes or pandemic situations leading to delays	8
R4	Difficulty in adjusting our schedule to that of the customer	9
R9	Coding is not done according to the design	10
R11	Imperfect sprint planning	11
R2	Complications in finding the right person for task allocation	12
R12	Mistakes due to poor standards in testing and documentation	13
R6	Handovers to a new team member due to personnel turnover can be problematic	14
R7	Change requests during the sprint execution	15
R8	Adaptation of agile methodology in the project if using it for the first time	16

5. Findings and discussion

Table 19 lists the risks identified in our previous experiments on the basis of their level of priority. Risks of the natures ‘challenges’, ‘difficulties’, ‘mistakes’, and ‘changes’ bagged the priority vector from highest to lowest order, respectively. Among the 16 risk factors identified under the aforementioned risk natures, R14 (proper utilization of funds) is found to be the most important risk factor, followed by R1 (complicated tasks) and R16 (convincing the customer with alternative technologies). R14 and R16 belong to the risk nature ‘challenges’, and R1 belongs to the risk nature ’difficulties’. Risk factors R6 (handovers to a new team member due to personnel turnover can be problematic), R7 (change requests during the sprint execution), and R8 (adaptation of agile methodology in the project if using it for the first time) occupy the last three ranking positions. All three risk factors belong to the ‘changes’ risk nature. This analysis is also verified for consistency using the Consistency Index, and the Consistency Ratio is found to be less than 0.1 for all the calculations, which acknowledges the result to be consistent. Thus, we find AHP an efficient tool for prioritizing risk factors during software risk management, especially in small and medium enterprises.

6. Limitations and threats to validity

In spite of the contributions, the study is susceptible to certain limitations and threats to validity.

Regarding construct validity, the risk factors used in this study were the outputs of our previous risk identification phase experiments conducted by unstructured and structured brainstorming sessions by software practitioners. The risks were assessed by experts in the relevant field and found significant by their qualitative ratings. Experts in this study performed the assessment by making pairwise comparisons, thus assisting in their prioritization.

The study also poses a certain amount of threat to external validity due to the shortage of generalizability. This is due to the fact that the responses were obtained from a limited number of experts in Kanyakumari district, India.

Another threat pertains to the AHP method of prioritization. The prioritization of the risk factors used in this study was purely based on the choice of the experts involved. Hence, any hasty decisions made by them are a threat to the outcome of the study, despite the achievement of internal validity with sufficient CR values at each stage.

7. Conclusion

In this paper, risks of four different natures named ‘difficulties’, ‘changes’, ‘’mistakes’, and ‘challenges’ identified through our previous experiments in software projects of small and medium enterprises were prioritized according

to the Analytic Hierarchy Process (AHP) methodology. The results of the study ensure statistically significant consistency. This prioritization helps in knowing the most and least significant risks, thus assisting in the upcoming risk mitigation activities. Thus, AHP is a noteworthy tool for making decisions with multiple criteria across several domains, including software risk management.

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